BIOGRAPHICAL SKETCH

NAME: Carstensen, Marcus

POSITION TITLE: Co-Founder and Chief Scientific Officer at OptoCeutics ApS

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EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Technical University of Denmark	B.Sc.	08/2016	Physics
Rensselaer Polytechnic Institute **(Exchange Scholarship)**	B.Sc.	08/2016	Physics
Technical University of Denmark	M.Sc.	08/2018	Physics
University of California, Berkeley **(Visiting Scholarship)**	M.Sc.	08/2018	Neurophotonics
Technical University of Denmark	Ph.D.	10/2023	Neurophotonics

A. Personal Statement

As the Co-Founder and Chief Scientific Officer at Optoceutics, I lead our clinical, regulatory, and intellectual property initiatives. The journey of Optoceutics began during my graduate studies at the Technical University of Denmark, where I co-founded the startup with a vision to revolutionize the treatment of Alzheimer's disease. My drive is fueled by a profound sense of purpose, relentless personal development, and a deep appreciation for teamwork. I firmly believe that the success of any project is built on the strength and collaboration of the team. Finally, I am an energetic and motivated individual with an insatiable curiosity for knowledge, ranging from the fundamentals of mathematics and physics to the latest advancements in nature and medical science. This diverse interest spectrum not only enriches my professional endeavors but also propels my passion for innovation and excellence. I thus give myself permission to take my time learning in-depth information while relishing the feeling of emerging from the abyss of detail to experience the wider entrepreneurial journey.

B. About

‣ Ph.D. from DTU Electrical Engineering (Photonics)

‣ Co-founder and CSO of OptoCeutics: A DTU and UC Berkeley spinoff

‣ Background in physics and engineering from DTU

‣ Published BSc. thesis in a peer-reviewed journal on high-speed holographic resonant later printing. The research project and technology later won the European Innovation Radar prize in Excellent Science, 2018

‣ Research assistant in experimental optics and electro-optics at DTU Nanotech and Fotonik during bachelor and graduate studies

‣ Awarded $3000 for best MSc. thesis and received around 1 million DKK in soft funding allocated to the research project

▸ Co-authored 10+ journal and conference papers

▸ Awarded a Visiting Scholarship at CITRIS, UC Berkeley, California

▸ Studied physics at Rensselaer Polytechnic Institute, USA, as part of my bachelor's degree in 2015

▸ Teaching assistant at the university level in mathematics @DTU; volunteer at UNF for mathematics and data science camps

▸ Principal investigator for Optoceutics within the large Marie Skłodowska-Curie ITN EU Project, ASTROTECH, funded by Horizon 2020

▸ Co-supervisor to three Ph.D. students

▸ Secured over 7 million DKK in soft funding; raised over 30 million DKK in equity

▸ Lead on four patent families, co-writing three

▸ Five years of experience in regulatory and clinical affairs for medical device development under MDR 2017/745

B. Positions and Honors

Positions and Employment

2014-2016	Teaching Assistant in "Advanced Engineering Mathematics 1", Department of Applied Mathematics and Computer Science, Technical University of Denmark
2016-2017	Research Lab Assistant, Department of Nanotechnology, Technical University of Denmark
2018-2019	Research Assistant in an Optics and Photonics Laboratory, Department of Photonics Engineering, Technical University of Denmark
2018-2023	Ph.D., DTU Photonics Engineering, investigating photonics technologies for the treatment and diagnostics of neurogenerative diseases
2018-2024	Co-founder and CTO at OptoCeutics, Denmark
2024-present	Co-founder and CSO at Optoceutics, Denmark

Honors

2018	DTU Green Challenge, Best MSc Thesis (3000$ USD personal price)
2018	Technology contribution to European Innovation Radar prize in Excellent Science

C. Contribution to Science

Google Scholar: https://scholar.google.com/citations?hl=en&user=QpZBqwkAAAAJ&view_op=list_works

Additionally, following prints within physics and photonics engineering:

BSc Thesis *(Feb 2016 – Jul 2016)*: **TITLE: Ultrafast Plasmonic Laser Printing** Technical University of Denmark (DTU)	**MSc Thesis** *(Feb 2018 – Aug 2018)*: **TITLE: Photonics Technologies and Innovation for Health Care** Technical University of Denmark (DTU)